

(Continued from *NOWELE* vol. 6)

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THE FATE OF THE NUMERAL '4' IN GERMANIC (2)

1.5. *The Gothic and West Germanic inflection*

1.5.0. In North Germanic, '4' inflects like a strong adjective (and accordingly cannot occur uninflected), whereas the higher cardinals are indeclinable. In Gothic and West Germanic, on the other hand, inflected forms of '4' occur only in certain circumstances (see §1.5.2.0., below), and are formed by adding *i*-stem nominal endings to the reflex of the form **feōwōr*. Moreover, '4' shares this peculiarity with the numerals from '5' to '19'. A further oddity is the fact that in Gothic only the dative and genitive add endings (there are no *+fidworeis*, *+fidworins*), whereas West Germanic has a full paradigm.

REMARK. In Old Saxon, the absence of inflected forms of '13-19' (duly noted by Gallée-Lochner 1910:234-235, Kieckers 1928:175, Szemerényi 1960:40) is a mere accident of transmission; these numerals are not found at all in the *Heliand* or *Genesis* (and, to my knowledge, do not occur inflected elsewhere).

Compared to the original Indo-European situation, where '4' was an *r*-stem plurale tantum (cf. §1.0.), both types of inflection found in the Germanic languages are clearly secondary. What is not clear, in the first instance, is the relationship of the two types of Germanic inflection one to another, and that of both to the paradigm of the parent language. These difficult questions will be addressed below

(§1.6.). For the time being, I shall merely describe the Gothic and West Germanic inflection.

1.5.1. The partial inflection of Gothic produces the following paradigm:

nom.-acc.	<i>fidwōr</i>
gen.	<i>fidwōrē*</i>
dat.	<i>fidwōrim</i>

The endings of the West Germanic languages are:

Old High German

	masc./fem.	nt.
nom.-acc.	<i>fiori</i>	<i>fioriu, -u</i>
gen.		<i>fioreo</i>
dat.		<i>fiorim</i>

Old Saxon only attests masc./fem. nom.-acc. *fiori* and dat. *fiuuarium*.

Old English

nom.-acc.	<i>fēow(e)re</i>
gen.	<i>fēow(e)ra</i>
dat.	<i>fēow(e)rum</i>

REMARK. A neuter form of *fēower* with the ending *-u* (*-o*), cited or implied in handbooks, does not seem to occur in standard Old English. The few examples I have noted are all late and dubious, being from tenth century Northumbrian texts (references here and elsewhere follow the system of *A Microfiche Concordance to Old English*): e.g. LiEpis (Skeat) 18 *in tal t̃ in regel ðone forma gegeadriges t̃ efnesæcgas feуро* < final *-o* of *feуро* in red ink > [In canone primo concordant quattuor mattheus marcus lucas iohannes]; JnGl(Li) 4/35 *feуро*

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monedō [quattuor menses]; DurRitGl (Thom-Lind) 47/14 [MS, Brown et al. 1969, 23r/20-22] *bif'a feoro netno* [ante quattuor animalia] ... *hundrað feortig feoro ðusendo* [centum quadraginta quattuor milia]; DurRitGl (Thom-Lind) 104/7 [MS 50r/3] *feor^o feortigv' hund' ðvsenda* [CXLIII milia] - DurRitGl also has a number of truncated forms of the numeral '4', e.g. <feor'>, presumably representing *feoro*, etc.

Compare Girvan's assessment of the situation of the numerals from '4' to '12' in general (1931:333): 'Het neutrum heeft misschien oorspronkelijk -u, -o gehad, later schijnt het echter den uitgang -e te hebben aangenomen, en wordt dan gelijk aan het masc. en fem.; in het l. north. is -o algemeen; deze uitgang is hier echter niet tot het ntr. beperkt'. I doubt very much that IONh -a, -o, -u is the etymological neuter ending; it is much more likely to be a 'general purpose' final vowel - perhaps influenced by the *ēo*-diphthong of the initial syllable?

It seems to me that the lack of a *+fēoweru* form could be either a syntactic or a morphological phenomenon: syntactic, if it is a question of absence of inflection in a further environment (cf. §1.6.4.5.); morphological, if it is the case that the neuter form is *fēower* from a putative earlier **fēoweru* (cf. §1.6.5.2.), with loss of final **-u*. (However, I have not noted convincing examples of inflected neuter forms of the cardinals *fif*, *syx*, *seofon*, or *nigon* either.)

Attestations and examples of the use of inflected forms are given in §1.5.2. The origin and development of these paradigms will be discussed in §1.6.

1.5.2.0. The generally accepted rule for Gothic and West Germanic is that, when used adjectivally and immediately preceding the qualified noun, the numerals are uninflected, otherwise - in adjectival function not immediately preceding the noun, or when used substantively - they are inflected. Compare Behaghel 1917:157 on the conditions for the appearance of inflected forms in Gothic:

1. sie erscheinen nur im genitiv und dativ, niemals im nominativ und accusativ.
2. sie erscheinen - mit einer ausnahme - nur in den fällen, wo das zahlwort nicht adjectivisch unmittelbar vor seinem substantiv steht; in diesen fällen aber sind sie ausnahmslose regel.

(The exception mentioned by Behaghel is the example of '9' cited below, but further down the same page he decides it is not an exception after all!)

As noted above §§1.5.0., 1.5.1., the West Germanic languages also have inflected nominative-accusative forms.

1.5.2.1. GOTHIC attests the following inflected forms of the numerals '4' to '19':

'4'

usliþan ... hafanana fram fidwōrim Mk 2/3

'a paralytic carried by four (men)'

(cf. *af fidwōr windam* Mk 13/27 'from the four winds')

'9'

in niuntēhundis jah niunē garaihtaizē L 15/7

'on account of ninety-nine [of] righteous (ones)'

Here, I take the numeral as a substantive (governed by *in*), with *garaihtaizē* a dependent partitive genitive.

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'11'

þaim ainlibim ataugida Mk 16/14

'showed to the eleven'

ataugids ist ... þaim ainlibim 1Kor 15/5

'is shown ... to the eleven'

'12'

qaþ ... du þaim twalibim J 6/67

'said to the twelve'

mip þaim twalibim Mk 4/10, 11/11

'with the twelve'

ains/ sums (...) þizē twalibē J 6/71, Mk 14/10. 43

'one of the twelve'

wintriwē twalibē L 8/42

'of twelve winters'

jērē twalibē Mk 5/42

'of twelve years'

'15'

swaswē ana spaurdim fimftaihunim J 11/18

'about fifteen furlongs'

Compare nominative *þai taihun* Mk 10/41 'the ten', and accusative *dagōs ahtau* L 2/21 'eight days', instances where we would have inflection if the case were genitive or dative. Unless we are to allow it as an exception, we would have to take the numeral in the following example as standing in apposition to the noun. Being thus in the nominative, *fidwōr* would regularly lack inflection:

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widuwō jērē ahtautehund jah fidwōr L 2/37

'a widow, (?one) eighty-four [in respect] of years'

Gk. χήρα ὥς ἑτῶν ὀγδοήκοντα τεσσαρῶν

1.5.2.2. I give here a sample of the use of inflected and uninflected forms of '4' in WEST GERMANIC.

(1) In OLD SAXON, we have: Hel(C) *sia (/sea) fiori* nom. pl. 9, 16, 32, (M) *mid them fiuuarun* dat. pl. 1190, beside uninflected *fiuuuar naht* acc. pl. 4084, 4131.

There is one 'irregularly' uninflected form in substantival use with a partitive genitive: 513 *Tha uuas siu uuiduuue ... fiuuuar endi hunahlude uuintro* (S, correspondingly M, C) though, as in the Gothic instance, we have here a complex hundred.

Both Old High German and English record divergences from the rule given in §1.5.2.0.

(2) OLD HIGH GERMAN has two 'irregularly' inflected forms, directly preceding the noun in adjectival function. (Square brackets indicate editorial additions.)

B 222/8 *sin kaleran andre* [feo]ri [lec]zun

legantur ali(a)e quatuor lectiones

B 248/21 *so qhvedames ... keleranem feorim ... fimfim pletirun*

ut diximus et lectis quattuor aut quinque foliis

These examples are early, but they both occur in the same text, which is, moreover, an interlinear gloss.

We may note that in

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MH 18.1.1. *so driror feorim kazokan ist stunton tak za habande*
sic ter quaternis trahitur horis dies ad vesperum

the numeral is inflected in adjectival function not directly preceding the noun (*stunton*), and so abides by 'the rules'. It is fairly clear that the glossator is slavishly following his Latin original, however.

Otherwise, the rule given above is observed in Old High German. Thus, uninflected directly preceding a noun: *fior taga* T 135/9, *fon feor uuinton* T 145/19; inflected, a) following a noun: *thie scriptora fiari* O 3.14.3, b) in substantival function: *in fieru sie iz gideiltun* O 4.28.2, *dero sint fieriv* Nb 219.17, *thiu uuas uuituua unzan fiору inti ahtuzug iaro* T 7/9. See further *Althochdeutsches Wörterbuch* s.v. *fior*. (In Middle High German, the numerals adopted adjectival genitive plural forms, before becoming flectionless - apart from occasional dative forms, cf. Wilmanns 1909:§211.9.)

(3) OLD ENGLISH has by far the richest attestation of the numeral '4' of all the West Germanic languages. This may be the reason why it records numerous exceptions to the supposed Gothic and West Germanic rule (above §1.5.2.0.). From an early date, there are instances of the uninflected numeral in substantival use (compare §1.5.1. REMARK, above); inflected forms in adjectival function immediately preceding the noun also occur. (The system of references follows that of *A Microfiche Concordance to Old English*.)

For examples of 'regular' uninflected adjectival use immediately preceding the noun, compare the following early examples: Ch 1482 (Harm D 2) [833-839] 68 *feower weðras*, ChronA (Plummer) 882 *ælfred cyning gefeaht wið feower sciphlēstas Deniscra monna*, Or (Bately) 132/26-27 *þa feower onwealdas þara feower heafedrica þisse middangeardes*, CP(H) 171/5 *mid ðam feower godspellum*. (N.B. If a metrical pause separates adjective and noun, inflection is considered 'regular', e.g. Rid 36/3 *Hæfde feowere . fet under wombe*, and the

example from ChristC given in the next paragraph - cf. Campbell 1959:284 fn3.)

For inflected adjectival use not immediately preceding the noun, compare Mart 5 (Herzfeld-Binz) 1698 [AU 12/A/6] *hwæt þa halgan godspelleras feowere sægdon*, ChristC 878 *Þonne from feowerum . foldan sceatum*. The readings of the two main manuscripts of GDPref 2/96/1 provide an interesting contrast of forms, and one that is completely in keeping with the 'rules': (C) *ac þa feawa, þe ic secge, ic ongæt æt feowrum his gingrum*, (H) *ac þa feawa, þe ic cyðe, ic oncneow æt his feower gingrum* [pauco quae narro quatuor discipulis illius referentibus agnovi].

For inflected forms when substantival, compare: Or(Bately) 107/18-19 *Hannibal ofpleah feowera sum* (see also the example from VercHom 1 given below); Ch 1507 (Harm D 11) 84 *minre yldstan dehter þære medemestan þære gingstan Ealhswiðe, him feowrum feower hund punda, ælcum an hund punda*; ÆInstSig18/1-2 *On hu fela gesceaftum stent þes middan eard? On feowrum: on fyre on lyfte, on wætere on eorþan*.

By far the largest group of exceptions is constituted by instances of uninflected substantival use. For example, *on feower* (*todælan* or *-ceorfan*, etc.) as HomS 24 (VercHom 1) 178-279 *þa todældon hie his hrægl on feower, ælcum hiora feowora his dæl* (compare the Old High German example from Otfrid cited above - **(to-)dailijan* takes the accusative). N.B. ÆGram. 7/7-11 *hwilon byð þæt stæfgefeg on anum stæfe, hwilon on twam ... hwilon on þrim stafum: [arx] wighus; hwilon on feower: [pars] dæl; hwilon on fif: [stans] standende; hwilon on syx: [stirps] stybb oððe mægo*; ByrM 1 (Crawford) 54/22 *twa oððe þreo, feower oððe fife oððe syxe*. (This last example suggests that inflected forms of *fēower* were avoided perhaps because the basic form was already disyllabic. Impressionistically, the same seems to hold for the other disyllabic cardinals *seofon* and *nigon*. However, a complete and reliable collection of data on the inflection of the numerals in Old

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English is the necessary basis for sound conclusions - as in all historical syntactic work. It is not possible to remedy this lack in the context of the present discussion.)

Inflected forms immediately preceding the noun in adjectival function are found in later texts, thus: *ÆCHom* I, 38 586/7 *nu wille we eow secgan þa getacnunge þæra feowera apostola namana*, *ByrM* 1 (Crawford) 10.2 *mid feowrum gesceaftum*.

(4) Inflected forms are relatively rare in OLD FRISIAN. In substantival function, I know only examples in the genitive and dative: e.g. (genitive) *therra fiowera aller elck en pund* 'to each of all the four a pound' *S* 493/5, *thera fiowera en* 'one of the four' (Schulzenrecht, quoted from Steller 1928:109/18), and also in the idiom *fiuw(e)ra sum* 'one of four'; (dative) *Djt is aerste fan dae fiowerum* 'this is the first of the four (sc. rights)' (*J* II, 352, quoted from Buma-Ebel 1977), *op wz fiouwren* 'between us four' *O* II, 99/14, *by fjouweren* 'amongst the four' *O* III 7/4 (quoted from Sipma 1927-1941). The absence of inflected forms in adjectival function is perhaps connected with the placement of the numeral immediately before the noun; compare *fiuwer etha* 'four oaths' *R* 83/12, *et thera fiuwer hernana hwelik* 'at each of the four corners' *R* 77/31, *mith fiuwer modernmegan* 'with four maternal relatives' *R* 53/27. The following is a possible example of an inflected adjectival form immediately preceding the noun: *Halssleeck twiska wede ende swaerde fyowera finghera breed fiower scillinga* 'a blow in the neck between garment and scalp four fingers wide: four shillings' (but it is conceivable that *breed* is a form of the noun *brede* 'breadth, width' rather than the adjective, and that we must accordingly take the construction as 'the width of four [of] fingers') *A* 245/24 (Codex Aysma, Bodleian Junius 78). (For the unspecified quotations, references are to the edition of Old Frisian texts by von Richthofen 1840, cf. vol. II of the reprint, page preceding 583.)

1.6. The interrelation of the North Germanic and Gothic-West Germanic paradigms and the question of their origin

1.6.0. We must try and clarify the historical relationship between the North Germanic paradigm, based on a stem-form **feður-* and following the inflection of the strong adjectives, and the paradigm of Gothic and West Germanic, which adds *i*-stem nominal endings to a stem **feðwōr*. This means reconstructing a Proto-Germanic state of affairs which is capable of accounting for the two types of Germanic inflection, while at the same time being a plausible reflex of the paradigm of the parent language (cf. §1.0.).

This is by no means easy, for the evidence of the two types seems to point in opposite directions. On the one hand, the North Germanic stem-form **feður-* invites comparison with the Old Indic oblique stem *catúr-* (cf. §1.4.2.), and this suggests the presence of inflection in Proto-Germanic. The Gothic and West Germanic situation, on the other hand, appears to indicate that only the fossilized Indo-European neuter plural form was inherited into Germanic, and that this flectionless form secondarily acquired *i*-stem flection.

This ambivalence is echoed in the literature. In so far as the issue is addressed at all, we find two opposing views on the question of the history of the inflection of the cardinal '4' in Germanic. Some scholars state that it was already, or became, flectionless at the Proto-Germanic stage, and then secondarily assumed flection (of divergent types) in the separate dialects. This view is held by, for example, Boer 1921:232-233, Krahe 1961:36-37, Markey 1976:46. Others have taken the view - explicitly, or by implication - that Proto-Germanic kept at least some degree of inflection in this numeral. Independently in both Gothic and West Germanic, it then became flectionless and later acquired *i*-stem forms; in North Germanic, however, the inflection was retained and subsequently remodelled. Compare Loewe 1902:75-77 (who sees a Gothic-West Germanic isogloss here, however, cf. §1.6.

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1.3.), Kluge 1913:252, Kieckers 1928:175 fn1, Schwarz 1951:99, Krahe-Meid 1969:89, Nielsen 1981:185.

1.6.1.1. The standard explanation of the origin of the Gothic and West Germanic inflection starts from an indeclinable **fedwōr*. It was first proposed by Osthoff (1878:131):

Die im gotischen belegten dativformen *taihun-i-m*, *fidvor-i-m*, sowie *ainlib-i-m*, *tvalib-i-m* sind ohne allen zweifel sämtlich als analogiebildungen von *þr-i-m* ausgehend.

(N.B. *taihunim* is attested only in *fimftaihunim* J 11/18, cited above §1.5.2.1.)

Compare also e.g. Loewe 1902:82, Szemerényi 1960:40. This explanation is taken as applying to West Germanic as well. I must point out that the development cannot be formulated in terms of an analogical proportion, but must be a type of contamination (cf. van Helten's objection to Osthoff's explanation 1905-1906:85). Nonetheless, '3' is the only reasonable source for the *i*-stem inflection of Gothic and West Germanic.

It is instructive to depict the process of the development in the form of a diagram: *þrim* → *fidwōrim* → *fimfim** etc. (cf. Szemerényi 1960:40). It seems very probable that the new dative forms spread from cardinal to cardinal in just this way (cf. below §1.6.4.2.). This behaviour is presumably connected with the peculiar property numerals have of standing 'at the same time in paradigmatic and syntagmatic relations: paradigmatic, since they share the same environment where they are in contrast; and syntagmatic since they commonly occur together in the act of counting' (Bazell 1977:40). Gothic would appear to have later extended the inflection to the genitive.

REMARK. The form of the genitive in Gothic agrees with the *i*-stem nouns, as *gastē*, rather than with '3', *prijē*. We thus have *twalibē*, not *+twalibijē*. This could be taken as additional evidence that the dative was the first form to acquire inflection. Once *fidwōrim* etc. had been created, the numerals were open to influence from the substantives, so Loewe 1902:82-83, Kieckers 1928:175-176. (Old High German attests the expected *i*-stem form of the genitive in both the noun and the numeral, cf. the paradigm given in §1.5.1.) See further §§1.6.4.1., 1.6.5.1. REMARK.

This theory apparently provides a neat explanation of both why the inflection is according to the *i*-declension, and why it is limited to the dative and genitive in Gothic. The further extension of the inflection in West Germanic can be attributed to the general tendency to fill out a paradigm.

1.6.1.2. A functional impetus might be inferred behind the adoption of this secondary inflection. Presumably, some kind of morphological marking of the syntactic relationship of the oblique cases was felt to be useful in the environments in which inflection occurs (cf. §1.5.2.0.). This circumstance would help explain the original adoption of a DATIVE ending from the cardinal '3'. The lack of inflection in adjectival function when directly preceding the headword can thus be explained by the fact that the syntactic relations were signalled clearly by the noun.

1.6.1.3. This shared innovation has been cited as a Gothic-West Germanic isogloss (Loewe 1902). I do not favour this interpretation, however, because there is no other evidence that points to a particularly close relationship between Gothic and West Germanic. (The partial agreement of Gothic and West Germanic over against North Germanic in the formation of the decads from '70' on results, of

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course, from innovation in North Germanic, cf. Szemerényi 1960:27, 43.) As so specific an innovation as the spread of the *i*-stem dative ending from the numeral '3' is less likely to have occurred twice than once, it would seem preferable to place the contamination in the Proto-Germanic period (cf. further §1.6.4.).

1.6.2.0. The evidence that the cardinal '4' retained some degree of inflection in Proto-Germanic comes from North Germanic. It is of both a functional and a formal nature.

1.6.2.1. Firstly, transfer from a (relic) inflectional pattern to another, highly productive, type is a more likely occurrence in a language undergoing general morphological simplification, than that a previously uninflected form should adopt inflection. All the more so, in the present instance, since the numerals from '5-19' remain doggedly uninflected in North Germanic, and would have provided strong support for an already uninflected '4'.

Another indication of the archaic nature of the North Germanic state of affairs is that, with '4' inflected, but '5-19' not, it is the same as that of Proto-Indo-European, cf. §1.2.1. (On the use of the masculine as the basic form, as in Indo-European, cf. §1.6.5.3.)

1.6.2.2. The second, and in my view crucial, argument is the fact that only the stem-form **feður-* will satisfy the North Germanic sound-laws. The long **-ō-* of a putative stem-form **feðwōr-* would not undergo syncope, but merely shortening, to *-a-*, cf. Krause 1948:§55. This would produce forms such as OIc. masc. *+fjǫðvarir*. The only sensible etymology of NGmc. **feður-* is the Indo-European oblique stem-form **k^wetúr-*.

REMARK. It is possible to interpret the North Germanic stem-form **feður-* as the combining-form (< PIE **k^wétur-*), which has somehow

replaced the inherited stem-form of the cardinal. PIE **k^wétur-* would yield PGmc. **fepur-*. (Go. *fidur-* has eliminated Verner's Law by adopting the *-d-* of the cardinal, cf. §2 below.)

Apparently, the combining-form **fepur-* would meet the formal requirements as well as does a putative reflex of Indo-European oblique-stem **feður-* (cf. §1.4.2. above). The medial **-þ-* of the combining-form can be assumed to have been voiced to **-ð-* medially in time to undergo the assimilatory change to **-γ-* (above, §1.4.3.).

But assuming that a Proto-North Germanic stem *þ*feðwōr-* (furnished with strong adjective endings) was replaced by the combining-form of the numeral does not make for a convincing explanation - there is no motivation for such a bizarre replacement.

1.6.3. However, if the North Germanic inflection is inherited, this causes grave chronological and stemmatic problems for the explanation of the Gothic and West Germanic paradigms given above (§1.6.1.). We would have to assume that Gothic, the earliest Germanic language of which we have extensive records, has already lost all trace of the Proto-Germanic inflection, and that West Germanic has given up the inflection quite independently, for it must have been still preserved at the North-West Germanic stage.

1.6.4.0. When we put it like this, the way out of the apparent impasse seems obvious: we must jettison the (in my view, quite unnecessary) assumption that '4' was flectionless when it was contaminated with the dative of '3' in Proto-Germanic (§1.6.1.3.). It seems to me just as likely, or - rather - more so, that such a development would take place if '4' itself had a dative form which was liable to contamination in the first place.

REMARK. We may note that just such an unmotivated contamination of a form of '4', by the equivalent form of '3', is evidenced by the

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earliest attested form of the ordinal '4th' in Indo-European, Vedic *turīya-*, Av. *tūriya-*. This reflects earlier $*(k^w)tur-i\grave{o}-$ for unattested 'original' $*(k^w)tur-ó-$, by contamination with $**tri(i)-\acute{o}-$ '3d' (cf. Szemerényi 1960:80-83, 92). Cf. further §3.

Further, all the Germanic dialects agree to the extent of having some kind of inflection in the cardinal '4', so we can provide a simpler account of the Germanic development, requiring fewer intermediate stages and less time-depth, if we posit a degree of inflection for Proto-Germanic.

Accordingly, I shall now present a scenario of the development of the inflection of the cardinal '4' in Germanic, which tries to integrate the various pieces of evidence into the simplest model. I must point out that everything said here is of necessity speculative, and is therefore advanced tentatively.

1.6.4.1. The formal reflex ($*fe\grave{d}ur-$) of the Indo-European oblique stem-form ($*k^wetúr-$) in North Germanic suggests that the cardinal '4' retained some vestiges of its Indo-European inflection in Proto-Germanic. As pointed out above (§1.2.1.), Go. *fidwōr* can only reflect the Indo-European neuter form; though WGmc. **fewwar* could be a regular reflex of $*k^wetuóres$, the Indo-European masculine form, economy dictates that we derive this from the same form as is attested in Gothic (cf. §1.2.3.). Like all the other Indo-European languages except Indo-Iranian and Celtic, Germanic has no trace of the special Indo-European feminine forms of the cardinals '3' and '4' (cf. §1.0.).

It would appear, therefore, that the original neuter nominative-accusative form and the masculine-neuter oblique stem-form (in the unvarying shape $*fe\grave{d}ur-$) were inherited by Germanic. As is borne out by Gothic, these forms were used for all genders in Proto-Germanic. Obviously, we must reconstruct the going set of case-forms - in the

plural only. If we put this information together, we arrive at the following probable (early) Proto-Germanic paradigm for '4':

nom.-acc.	<i>feðwōr</i>
gen.	<i>feðurōⁿ</i>
dat.	<i>feðurmiz</i>

It is difficult to determine the syntactic rules for Proto-Germanic. In view of the Indo-European situation, it seems most probable to me that the numeral was inflected in all environments (that is, even when immediately preceding the noun). It also seems easiest to arrive at the North Germanic situation from this starting-point. Alternatively, Proto-Germanic might already have developed the same usage as Gothic (no inflection of genitive and dative when immediately preceding the noun). Because the nominative-accusative form was 'ending-less', a situation retained in Gothic, the inflection will have only manifested itself in the genitive and dative, cf. §1.5.2.0. (To their ambiguous syntactic position, halfway between being nouns and adjectives, I attribute the fact that the numerals '2' to '4' did not adopt the inflection of the 'pronominal adjectives', along with all other Germanic adjectives when in strong inflection.)

1.6.4.2. If we now place the remodelling of the inherited dative of '4' (the result of contamination with the ending **-im(i)z* of **prim(i)z*, cf. §1.6.1.1.) within the Proto-Germanic period (§1.6.1.3.), we obtain a late Proto-Germanic dative form **feðurim(i)z*. The inflection cannot have spread to the higher cardinals at this stage, because it has not done so in North Germanic. The stagewise spread of the inflection posited above (§1.6.1.1.) would thus appear to be confirmed by this state of affairs. On the present interpretation, the limitation of inflection to '4' in North Germanic is quite simply the archaism it appears to be (cf. §§1.0., 1.5.0., 1.6.2.1.).

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This gives us the following probable set of forms for late Proto-Germanic:

nom.-acc.	* <i>feðwōr</i>
gen.	* <i>feðurōⁿ</i>
dat.	* <i>feðurinz</i>

1.6.4.3. This paradigm seems to me to offer three possibilities of further development as far as the stem-form is concerned:

- (a) no change;
- (b) levelling of the allomorph of the nominative-accusative throughout the paradigm;
- (c) levelling of the allomorph of the genitive and dative throughout the paradigm.

I would suggest that Gothic adopted possibility (b), whereas the paradigm was preserved unchanged in North West Germanic, (a); North Germanic subsequently realized option (c), and West Germanic, like Gothic, carried through alternative (b). (It seems possible to me that 'Crimean Gothic' *fyder* reflects the North-West-Germanic zero-grade stem-form **feður-*, unless it is borrowed from Bible Gothic *fidwōr*. This seems better than the suggestion that *fyder* is somehow the combining-form, Stearns 1978:136, or has been influenced by it, Grønvik 1983a:27-29.)

REMARK. As mentioned in §1.0. above, Indo-Iranian is the only branch to preserve the original paradigm more or less intact. Elsewhere, drastic simplification has taken place and new paradigms have been constructed from the ruins of the inherited one.

As noted above (§1.0.), Celtic is the only branch apart from Indo-Iranian to preserve the special feminine form. The original neuter

form has been lost, however. A new neuter form, which differs from the masculine only in that it lenites, has been created on the model of the numeral '3' (cf. Thurneysen 1946:§232.4).

In Greek, the feminine has been lost without trace. The original neuter nominative-accusative form does not survive anywhere. Most dialects have created a new paradigm based on the zero-grade stem-form, with a single masculine-feminine form (an apparent archaism) and a derived neuter. Thus we have Attic τέτταρες m./f., τέτταρα n., built on the alternant **-turC-* (Ionic τέσσερες is of course not ancient, but due to assimilation). Lesbian πέσσυρες (Hom. πίσυρες) presumably continues the alternant **-turV-*. West Greek τέτορες m./f., on the other hand, continues the Indo-European masculine form, **k^wetuóres*, with loss of the *-u-*.

In Italic, Latin has only the indeclinable form *quattuor* (in my opinion the fossilized neuter, cf. §1.2.1. REMARK., above). Corresponding to this, Oscan provides 'petora', apparently the neuter re-characterized, but also the mutilated form **pettiur** ['quattuor'] (Vetter 1953:102), apparently reflecting earlier **k^wetur-*. (The phonological development is *ũ > iũ* after a dental, with *i* later geminating the *t*, cf. Lejeune 1970:296-298.) Proto-Italic, like Proto-Germanic, thus seems to have preserved some degree of inflection, with different stem-forms being generalized in the daughter dialects.

The Lithuanian paradigm is built on the zero-grade stem-form **k^wetur-*, to which are added *ja*-stem adjective endings (presumably replacing earlier *i*-stem inflection, developed from the inherited *r*-stem inflection). Cf. also §1.6.5.3. REMARK.

1.6.4.4. On the present interpretation, the identical levellings of Gothic and West Germanic, and also the fact that these two languages extended the inflection beyond '4' to '19' (§§1.5.0., 1.6.1.1.), must be the result of chance. Obviously, this is a slight drawback.

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However, if we consider the levelling of the nominative-accusative stem-form into the genitive and dative of '4' to be the catalyst for the extension of the inflection to the other numerals, we only have to posit one chance innovation (out of a choice of two) in common. The relationship **feðwōr* : **feðwōrimz* provides a better model for **fimf* : **fimfinz* etc. than does **feðwōr* : **feðurimz*. (As mentioned above, §1.6.1.1., the creation of PGmc. **feðurim(i)z* in the first place was not the result of an analogical proportion.)

REMARK. The distinction between inflected '1' to '4' and indeclinable '5' to '19' has, not surprisingly, been lost or blurred in several of the Indo-European languages.

In Old Indic, the cardinals from '5' to '19' from an early date adopted inflection for case (in the oblique cases only).

In (East) Baltic, '5' to '9' have adopted the same inflection as '4'.

Conversely, in Latin, *quattuor* has become indeclinable, and thus joined '5' to '19'.

The distinction is intact in Old Irish, and, as we have seen, North Germanic; it survives fairly well in Greek.

1.6.4.5. The potentially more damaging Gothic-West Germanic agreement is in the syntactic rules for the use of inflection. Under the present interpretation, the explanation reported above (§1.6.1.2.) has to be more or less turned on its head.

How we envisage the development depends on our view of the syntactic rules of Proto-Germanic (cf. §1.6.4.1.). Gothic and West Germanic have either essentially retained the Proto-Germanic situation, or have abandoned inflection when '4' is adjectival and immediately preceding the noun. (At any rate, we are not now reckoning with a creation of inflection; cf. §1.6.4.0.)

If the genitive and dative were inflected when in adjectival function immediately preceding the noun in Proto-Germanic, then it

seems to me that we have in this situation the 'seeds' for the further syntactic development of the three Germanic branches. The genitive and dative will have been inflected in all environments, whereas the nominative-accusative form (the original neuter) was uncharacterized in all environments. It was thus a lop-sided morpho-syntactic system.

A resolution of this unbalanced situation seems possible either (a) by the introduction of inflection into all environments, or (b) by abandoning the inflection of the genitive and dative in adjectival function immediately preceding the noun (a step in the direction of 'group-inflection'). I suggest that we can posit a sequence of development analogous to that affecting the stem-form (§1.6.4.3., above). Gothic adopted option (b). North-West Germanic created endings for the nominative and accusative forms when they were not adjectival immediately preceding the noun (see §1.6.5.2., below), but otherwise preserved the Proto-Germanic state-of-affairs. North Germanic generalized the use of endings to all environments (a), whereas West Germanic mainly went the same way as Gothic (b) thus 'balancing' the system, but also both increased and decreased the domain of inflected forms (evidence mainly from Old English), cf. §1.5.2.2.

1.6.5.1. GOTHIC shows two morphological innovations vis-à-vis the late Proto-Germanic situation outlined above (§1.6.4.2.): levelling of the stem-form *fidwōr-* into the genitive and dative, and subsequent extension of inflection to the higher numerals.

REMARK. The question of the vowel-quality (*-ē*) aside, we may note that, on the present analysis, the form of the Gothic genitive ending is merely an archaism, and requires no special explanation. In Old High German, the form of the genitive plural is the result of the general re-modelling of the paradigm on that of the *i*-stems. (Cf. §1.6.1.1. REMARK above.)

1.6.5.2. It seems probable that in NORTH-WEST GERMANIC the original consonant-stem paradigm was wholly rebuilt according to the *i*-stem pattern under the influence of the contaminated dative *feðurimz*. The evidence of North Germanic and Old High German further suggests that a distinction between neuter and masculine/feminine was created in the nominative-accusative by re-marking the neuter with the ending of the strong adjectives. (Likewise, the creation of feminine forms for '3' on the model of the strong adjectives seems to belong to the North-West Germanic period. It is noteworthy that '3', like '4', has been influenced by the strong adjective inflection, even though the *i*-inflection was here 'built into the stem' as it were.) In uninflected use, however, the original uncharacterized form was retained for both neuter and masculine-feminine.

It seems, therefore, that we must reconstruct, beside uninflected **feðwōr*, **feðwōrīz* m./f. and **feðwōru* nt. as the North-West Germanic nominative forms, presumably adding a masc./fem. acc. **feðwōrinz*(?). Compare the innovated NWGmc. masc. acc. pl. **þrijanz*, beside inherited *þrins* of Gothic. Both these accusative forms were later lost in West Germanic as a result of nominative-accusative syncretism. It seems most likely that the apparent absence of an Old English reflex of the neuter form **feðwōru* is secondary, cf. §1.5.1. REMARK.

1.6.5.3. In NORTH GERMANIC, the zero-grade form of the stem has been levelled throughout the paradigm, and the inherited inflection has been completely replaced by strong adjective inflection (presumably a corollary of the syntactic development posited in §1.6.4.5.). As a result separate feminine forms have been created.

In North Germanic, *i*-stem adjectives regularly became *a-/ō-* adjectives (unless they had stem-final *g* or *k*, in which case they became *ja-/jō-* adjectives, cf. Noreen 1923: §§424.1, 431). We may

remember that the original *i*-stem form of the dative plural, *-im*, was replaced by *-um* at an early date (cf. YRu. **-gestumR** Stentoft), thus merging with the form of the strong adjectives.

The fact that the 'basic' form of the cardinal, used in counting etc., is the masculine *fjórir*, looks superficially like a significant archaism, cf. OInd. *catvāras*. On the present theory, this cannot be the case, however. Instead, I suggest that the newly-formed masculine of '4' was used in this way on the analogy of the cardinals '1-3'.

REMARK. The development of the cardinal '4' in North Germanic is thus typologically identical with its development in Lithuanian (cf. §1.6.4.3. REMARK, above). Like ON *fjórir*, Li. *keturì* is based on the generalized zero-grade stem-form, and has adopted the inflection of the strong adjectives (it inflects like an indefinite *ia*-stem).

Lithuanian differs from North Germanic, however, in having preserved an instance of the original consonantal inflection in the masculine accusative plural, *kēturis* < **k^wetúrns* (= OInd. *catúras*, §1.0.), though the expected adjectival ending, *kēturius*, is found in dialects, cf. Stang 1966:278. As a general rule in Lithuanian, consonantal inflection is replaced by *i*- (or *io*-/ *ia*- ~ *ie*-) inflection (cf. Stang 1966:219). We may especially note that the language has neither consonant- nor *i*-stem adjectives. Lithuanian has lost the neuter gender, and the attested feminine forms are innovations.

1.6.5.4. Apart from levelling the stem-form of the nominative-accusative plural throughout the paradigm and the syncretism of the nominative and accusative (plural), the WEST GERMANIC languages have essentially preserved the putative North-West Germanic paradigm. In Old High German, the neuter form has adopted the pronominal *-iu* ending. Like North Germanic, Old English generalized *-um* as dative plural for all stem-classes. (It has been suggested above, §1.5.1. REMARK, that in at least some instances, *fēower*, in a context where

neuter inflection might be expected, is the reflex of earlier NWGmc. *fēoweru.)

2. The combining-form

The combining-form of '4' inherited by Germanic had the pre-Verner's Law shape *fepur-. Whether this continues an Indo-European type *k^wetur(V)- (as in the Old Indic combining-form *catur-*) or has developed via *fepwur- from *k^wetur(C)- (as in Gk. (τε)τρα-) is for our purposes immaterial. At any rate, Germanic preserves no trace of the Indo-European metathesized (pre-consonantal) variant *k^wetru-, on which see Wackernagel 1881:283, Brugmann 1904:§§155 A.3, 442.4, also Wackernagel-Debrunner 1930:348, with references.

Whereas OSw. *fiaðr-*, *fioper-* *fiæper-* (cf. Kluge 1879b:393-394, argued by Sievers 1879:575-576) could reflect either IPGmc. *-p- or *-ð-, the Gothic and Old English continuants of the Proto-Germanic combining-form provide contradictory testimony as to the voicing of the medial dental in late Proto-Germanic (after the operation of Verner's Law). Compare, on the one hand, Go. *fidurdōgs*, *fidurfalps*, on the other, OE *feodur-*, *feodor-*, *feod̥er-*, *feðer-*; *fīper-*, *fyper-* (on the vocalism of these last two forms, cf. Ross 1974). The discrepancy was definitively discussed by Kluge (1879b:394-395), whose argument deserves to be quoted in full. After citing the combining-form found in Old and Middle English, he continues (394):

Wie aber ist die innere spirans gegenüber der got. media zu erklären? Nach dem grundsatz, dass man für gewöhnlich der schwierigeren form höhere altertümlichkeit gibt als der irgendwie durch secundäre einwirkungen, etwa durch analogiebildung erklärbaren form, müssen wir annehmen, dass im germ. composita mit 'vier' als erstem gliede dies in der gestalt *fīpur* besaßen. Dem ae. *fyðerfête* muss demnach

germ. *fīpurfôtuz* = vierfüssig entsprechen, obgleich wir für das got. nach *fidurdôgs*, *fidurfalps* u. s. w. ein *fidurfôtus* vermuten dürfen. Da aber die media des got. aus dem uncomponierten worte entlehnt sein kann, für das ae. eine ähnliche annahme unmöglich ist, kann nur ae. *fyðerfête* reflex einer germ. grundform sein. Die betonung des germ. *fīpurfôtuz*, die auf dem Vernerschen gesetz beruht, stimmt schön mit altind. *cātuṣpād* überein.

Kluge further remarked (1879b:395):

Vor der consequenz, die sich aus diesem nachweis ergibt, wonach *fēpur* als erstes glied im Bahuvrīhicompositum gegolten hat, wird niemand zurückschrecken: diese consequenz lautet: wo in der zusammensetzung die form mit innerer spirans in den einzelnen dialecten fehlt, ist sie durch das selbständig gebrauchte zahlwort verdrängt: so drang ja auch im bereich des engl. selber das einfache *feor* oder *fewer* ... in die zusammensetzungen ein und damit gieng das altertümliche und wertvolle *fyðer* zugrunde.

In fact, a reflex of PGmc. **fēpur-* is preserved to this day in the placename *Featherstone* (Nb, St [OE *feoperstan* Ch 1380 (Dugdale) 3], YW), beside *Fourstones* (Nb) with the linguistically later combining-form *fewer-*, both denoting 'tetraliths' (cf. Smith 1956:171).

The situation in Old Frisian is similar to that in later English, with the first elements of *fiuwerfald* and *fiārfōte*, corresponding to *fiuwer* and **feor* (on the latter, cf. Siebs 1901:1198, and §1.3., above).

Perhaps Lex Salica *fitter-* belongs here, (so Sievers 1879:576, cf. van Helten 1900:514-516). By the time of our records of Old High German at any rate, the reflex of original PGmc. **fēpur-* has been replaced by the form of the cardinal, thus the Old High German combining-form *feor-*, as in *feorfuozī*, *fiorfalt*.

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In North Germanic, the combining-form has a motley collection of forms (and spellings). These presumably represent varying phonological developments and analogies, at different stages, of PGmc. **feþur-*. (Influence from the various forms of the cardinal is probably not to be ruled out either.) Thus: *fiogor-* (< **feuþur-C-*); *fiaðr-*, *fioper-* etc. (< **feuþr-V-*, possibly with an intervening stage with analogically restored *-ur-*); *fjór-* (< **feuþr-*, analogically introduced before a consonant?); *fer-* must descend from a form without breaking, **feþr-* (which Noreen suggested was a weak-stress form, 1897:41, cf. 1923:§292) - this form is evidently continued in the 'distributive' *fernir* (pl. adj.), ultimately from something like **k^wet_uno-* (cf. Lat. *quaterni*).

Aside from the medial consonant *-d-*, the vowel *-u-* of the second syllable of the Gothic form *fidur-* requires a word. The question at issue is the absence of a 'breaking'-spelling <-aur-> indicating the lowering of *u* before *r* in Gothic, as is usual in root syllables. According to Hermann Paul (1894:334), pre-Gothic **-ur-* yielded Gothic **-ar-* (an unacceptable view) and *fidur-* is to be read as *fidūr-* (as de Saussure had supposed 1922:197 [=1879], reconstructing, in modern notation, *þ*^wetuxr-*, to *þ*^wetux-ór-*). Hamp 1980:40-41 has suggested a plausible way in which a long *-ū-* could have arisen, through the re-analysis of a Sievers' Law variant of the zero-grade stem-form **^wtu_ur-C* (beside **^wtur-V*) as **^wtuur-*, later revocalized. (Hamp uses this variant to explain the long **-ū-* found in the Slavic form of '4': OCS *četyre*.) But it seems to me that the best interpretation is, with Johannes Schmidt (1870:276, 280), to suppose that 'breaking' of *-u-* did not apply to unaccented syllables in Gothic.

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(To be continued in *NOWELE* vol. 8)